

CRR — Canonical Equations and Axiomatic Commitments

Coherence – Rupture – Regeneration

Synthesised post-campaign canonical reference. Equations and conventions consistent with *CRR_FINAL_CANONICAL.md*, *F_Structure.pdf* (Sabine 2025), and the corrections / extensions recorded across Sessions 1–15 of the campaign audit (*notes/conventions.md*, *notes/session_11_no_regulator_baseline.md*, *notes/session_12_pre_registrations.md*, *notes/session_14_literature_audit.md*).

Axiom numbering follows *F_Structure.pdf* §1.1. Convention dictionary (C1–C5) follows *notes/conventions.md*. Mathematical-claim numbering (M1–M22) follows *notes/decomposition.md*.

1. The six axioms

The canonical CRR axiom set, as committed in F_Structure.pdf §1.1 and reproduced verbatim:

Axiom 1 — Temporal Primacy

Process is ontologically prior to substance. Every entity at every scale is constituted by a temporal cycle of accumulation, transition, and reconstruction.

Axiom 2 — Coherence Accumulation

Any bounded system accumulates coherence over time:

$$\mathbf{C}(\mathbf{x}, t) = \int_0^t \mathbf{L}(\mathbf{x}, \tau) d\tau$$

where $\mathbf{L}(\mathbf{x}, \tau)$ is the local coherence rate. Under the M13 identification, $\mathbf{L}$ is the Fisher-Rao squared infinitesimal speed on the system's statistical manifold, so $\mathbf{C}$ is accumulated Fisher information.

Axiom 3 — Instantaneous Rupture

The transition between regimes is a Dirac delta $\delta(\text{now})$, scale-invariant: the same topology governs a synapse firing and a star collapsing.

Axiom 4 — Memory-Weighted Regeneration

After rupture, the system reconstructs via:

$$\mathbf{R}[\chi](\mathbf{x}, t) = \int_{-\infty}^t \phi(\mathbf{x}, \tau) \cdot \exp(\mathbf{C}(\mathbf{x}, \tau)/\Omega) \cdot \Theta(t - \tau) d\tau$$

where high-coherence past states are exponentially amplified, and Θ is the Heaviside causal cut.

Axiom 5 — Universal Rupture Condition

Rupture occurs when the saturation condition holds:

$$\mathbf{C} \cdot \Omega = 1$$

This is the Cramér-Rao bound saturated under the M13 identification: the system has extracted the maximum information available from its current configuration.

Axiom 6 — Symmetry Determines Ω

The system precision Ω is the inverse of the closed-geodesic length on the continual memory-bearing manifold:

$$\Omega = 1 / \phi_{\mathbf{G}}$$

For Z_2 (rupture-only, π geodesic): $\Omega = 1/\pi$. For $SO(2)$ (continuous-phase circle, 2π geodesic): $\Omega = 1/(2\pi)$.

2. The three core equations

CRR is a temporal grammar with three canonical operations on the coherence trajectory of a bounded system:

2.1 Coherence (the past)

$$C(x, t) = \int_0^t L(x, \tau) d\tau$$

with the canonical M13 identification: $L(x, \tau) = (d\theta/d\tau)^T G(\theta) (d\theta/d\tau)$, the Fisher-Rao squared speed on the statistical manifold parameterised by θ , with G the Fisher information matrix. Coherence is the squared arc-length action.

2.2 Rupture (the present)

$$\delta(\text{now}) \text{ when } C \cdot \Omega = 1$$

Three independent arguments force this rupture event to be structurally Z_2 (binary, Bernoulli($1/2$)):

- Dirac-delta form: δ has codomain $\{0, 1\}$.
- Heaviside-derivative form: cumulative-rupture-counting $N(t)$ has integer increments.
- Cramér-Rao saturation: at $C \cdot \Omega = 1$ the natural sufficient statistic is binary.

2.3 Regeneration (the future)

$$R[\chi](x, t) = \int_{-\infty}^t \phi(x, \tau) \cdot \exp(C(x, \tau)/\Omega) \cdot \Theta(t - \tau) d\tau$$

Three components: $\phi(x, \tau)$ the resource field; $\exp(C/\Omega)$ the coherence-weight (high- C past states amplified exponentially); $\Theta(t - \tau)$ the Heaviside causal cut. M14 identifies $\exp(C/\Omega)$ as the unique MaxEnt distribution under a mean-coherence constraint (Boltzmann-Gibbs, capped at $T1^*$).

M20 identifies $R[\chi]$ as the right Kan extension of the coherence-history functor along the rupture-inclusion functor.

2.4 The precision parameter Ω

$$\Omega = 1 / \phi_G$$

where G is a compact connected Lie group acting as the continual memory-bearing manifold and ϕ_G is its bi-invariant closed-geodesic length. Convention C4: Ω is *inverse* geodesic length — short geodesic $\Leftrightarrow$ high precision. Convention C5: $\Omega = \mu(A_{\text{coherent}})$ under Kac's lemma for ergodic systems.

3. Key derived identities

3.1 The $CV = \Omega/2$ prediction (M1)

$$CV = \text{std}(\tau_{\text{rupture}}) / E[\tau_{\text{rupture}}] = \Omega / 2$$

Bernoulli($\frac{1}{2}$) noise model at the rupture threshold gives $\text{std} = \frac{1}{2}$ and $\text{mean} = 1/\Omega$, so $CV = \Omega/2$. Independent of the phase manifold G ; the noise scale is intrinsic to Z_2 rupture.

3.2 Lie-group generalisation (M22)

$$CV_G = 1 / (2 \cdot \phi_G)$$

Predicted CV across canonical compact connected Lie groups:

G	ϕ_G	Ω_G	CV_G
Z_2 (rupture only)	π	$1/\pi$	$1/(2\pi) \approx 0.1592$
$U(1) \cong SO(2)$	2π	$1/(2\pi)$	$1/(4\pi) \approx 0.0796$
$SU(2) \cong S^3$	2π	$1/(2\pi)$	$1/(4\pi) \approx 0.0796$
$SO(3) = SU(2)/Z_2$	π	$1/\pi$	$1/(2\pi) \approx 0.1592$
T^2 (per generator)	2π	$1/(2\pi)$	$1/(4\pi) \approx 0.0796$
$SU(3)$	$2\pi\sqrt{3}$	$1/(2\pi\sqrt{3})$	$1/(4\pi\sqrt{3}) \approx 0.0459$

3.3 Cramér-Rao / Heisenberg-Gabor saturation (M3, M4, M5)

$$\text{Var}(\theta) \geq 1/I(\theta) \Leftrightarrow \Delta t^2 \cdot \Delta \omega^2 \geq 1 \Leftrightarrow C \cdot \Omega \geq 1$$

Three equivalent forms of the same uncertainty bound under (time $\leftrightarrow$ frequency, parameter $\leftrightarrow$ estimator) parameterisations. Saturation at $C \cdot \Omega = 1$ is the rupture condition.

3.4 The beauty function (M12)

$$B(C) = \exp(C/\Omega) \cdot (C^* - C)$$

Peaks at $C^* - \Omega$ by setting $dB/dC = 0$. Identifies the optimal coherence — the location of maximal agency / aesthetic tension — as just before the rupture threshold.

3.5 Anti-correlation under rupture composition (M11)

Two Z_2 rupture channels composing on a single $SO(2)$ phase manifold produce anti-correlation:

$$\rho = -1/2$$

Derived from variance-preservation $\text{Var}(X+Y) = \text{Var}(X)$ when X, Y are equal-variance Z_2 ruptures composing on shared $SO(2)$.

3.6 The α^3 embedding cost (F_Structure §6.2)

For atomic-scale transitions, where the rupture event (spontaneous emission) couples through 3D EM vacuum:

$$CV_{\text{atomic}} = \alpha^3 / (4\pi \cdot f)$$

with α the fine-structure constant and f a periodic-table-group-determined geometric factor. α^3 is the cost of embedding an $SO(2)$ coherence cycle in 3D EM space (one factor of α per spatial dimension).

Empirically validated across 49 elements (88% within 20%, median error 5.4%) and the alkali series Li → Fr (P15, T3 promotion).

3.7 The $\sqrt{2}$ precision-allocation ratio

Optimal Kelly / log-utility precision allocation between two channels:

$$\pi_p / \pi_s = \sqrt{2} \approx \mathbf{1.414}$$

Under the precision = 1/variance identification, this is the same algebraic identity as the variance-summation $\sqrt{2}$ for two i.i.d. equal-variance channels — the two derivations coincide. Empirically observed in the SILSO sunspot / Hale-cycle CV ratio (1.382, 2.3% from $\sqrt{2}$; Session 9 audit).

4. The CV ladder — substrate identification

Empirical / mathematical tier structure for inter-rupture-CV identifications, formalised post-campaign:

Tier	CV	Mechanism	Empirical examples
Deterministic limit	$\rightarrow 0$	rich G , perfect sync	atomic clocks
$SU(3)$ / higher	0.0459	richer compact group	(speculative)
$SO(2)$ / $SU(2)$ / T^2	0.0796	continuous closed-geodesic	solar Hale, GW BBH, intra-woman menstrual
Class B regulated	0.04 – 0.13	autonomous + feedback control	gait stride, hemispheric, Schwabe
Z_2 / $SO(3)$	0.1592	discrete binary phase	menstrual pooled, respiratory (Bull 2019)
Mid-regime	0.30 – 0.70	partial regulation	NBER recessions, Cascadia α , Vela glitches
No regulator	≈ 1.00	no phase manifold (Poisson)	cyber, Kp storms, UHECR, M-dwarf flares
Self-exciting (Hawkes)	> 1.30	event-triggered intensity	VIX spikes, SPX drawdowns (financial only)

Critical structural points (per Session 14 literature audit):

- **No-regulator boundary $CV = 1$** is the upper edge of the M22 framework ($G = \emptyset$, $\Omega \rightarrow 0$) and is the most empirically robust tier (5+ independent domains).
- **The Hawkes ($CV > 1$) regime appears human-social-system-specific.** Pulsar glitches, volcanic eruptions, and M-dwarf flares all FAILED Hawkes pre-registrations and instead sit at no-regulator. Hawkes requires explicit positive-feedback dynamics.
- **Mid-regime (0.3 – 0.7)** is the partially-regulated tier where neither autonomous Lie-group identification nor full no-regulator applies. Empirical examples cluster near $\sqrt{2}/2 \approx 0.71$ (boundary of two-channel precision-allocation).

5. Convention dictionary (notes/conventions.md)

C1 — Rupture is Z_2 by construction

Forced by three independent structural arguments (Dirac-delta form, Heaviside-derivative form, Cramér-Rao saturation under M3). The ' Z_2 ' label refers always to the rupture topology, not to a substrate alongside $SO(2)$.

C2 — Phase manifold is a compact connected Lie group

G is the continual memory-bearing manifold with bi-invariant Riemannian metric and closed-geodesic length ϕ_G . Canonical examples: $U(1) \cong SO(2)$, $SU(2) \cong S^3$, $SO(3) = SU(2)/Z_2$, T^n , $SU(3)$.

C3 — Two distinct Ω 's

$\Omega_{\text{geo}} = 1/\phi_G$ (geometric Ω of phase manifold). $\Omega_{\text{int}} = Z_2$ -intrinsic precision in normalised Bernoulli units = 1 by construction. The brief's ' $\exp(C/\Omega) \rightarrow e$ at $C \cdot \Omega = 1$ ' is correct in intrinsic units only. Default convention: Ω refers to Ω_{geo} .

C4 — Ω is inverse geodesic length

Large $\Omega \Leftrightarrow$ short geodesic $\Leftrightarrow$ high precision. On positively curved manifolds (Bonnet-Myers): $\Omega \geq \sqrt{\kappa/\pi}$ with equality on the round sphere. Brief's earlier ' $\Omega = \pi/\sqrt{\kappa}$ ' is an inversion typo (M16 corrected).

C5 — Ω is rate-like under Kac's lemma

For ergodic systems with coherent region $A \subset X$: $\Omega = \mu(A_{\text{coherent}})$, not $1/\mu(A)$. Brief's earlier inverse reading is a typo (M19 corrected).

6. The 22 mathematical claims (M1 – M22)

#	Claim
M1	$CV = \Omega/2$ with no free parameters
M2	$Z_2 : SO(2)$ CV ratio is exactly 2 from arc-to-ring topology
M3	$C \cdot \Omega = 1$ saturates the Cramér-Rao bound
M4	$C \cdot \Omega = 1$ saturates the Heisenberg-Gabor uncertainty
M5*	M3 and M4 are the same theorem (relabelling cap T1*)
M6	Fourier transform is the trivial CRR specialisation
M7	ϕ is the dominant eigenvalue of the depth-two regeneration operator
M8	Depth two is the minimum for KAM-stable ergodicity
M9	ϕ -rotated CRR has singular-continuous (Süt■-class) spectrum
M10	$1/\alpha = 137.0324$ fixed point of CRR self-consistency equation
M10-α^3	Subatomic CV scales as $\alpha^3 \times (8/3\pi)$ at leading Bethe order (T3)
M11	$Z_2 + Z_2 \rightarrow SO(2)$ composition gives $\rho = -1/2$
M12	Beauty $B(C) = \exp(C/\Omega) \cdot (C^* - C)$ peaks at $C^* - \Omega$
M13	C is identified with accumulated Fisher information $I(\theta)$
M14*	$\exp(C/\Omega)$ is the unique MaxEnt regeneration kernel (relabelling cap T1*)
M15	Z_n hierarchy: $CV = n/(4\pi)$ for cyclic Z_n
M16	Bonnet-Myers gives $\Omega \geq \sqrt{\kappa/\pi}$ on positively curved statistical manifolds
M17	C is quadratic variation $[\mu, \mu]_t$ in martingale formulation

M18	Rupture time τ_{Ω} is an optimal stopping time (SPRT)
M19	Poincaré + Kac make rupture inevitable for ergodic systems
M20	$R[\chi]$ is the right Kan extension of coherence-history along rupture-inclusion
M21	$C \cdot \Omega = 1$ saturates Cramér-Rao + Heisenberg-Gabor + TUR
M22	$CV_G = 1/(2 \cdot \phi_G)$ for any compact connected Lie group G

Tier-equivalent conventions: T0 = speculation; T1 = derivation consistent; T1* = canonical-result relabelling cap; T2 = empirical consistency; T3 = pre-registered prediction confirmed; T4 = independent unaffiliated replication. Campaign-current campaign T3 count: M10- α^3 , P15. T4 count: 0.

7. The two- Ω disambiguation (per C3)

Ω in CRR refers to two distinct quantities that should be disambiguated:

Symbol	Definition	Domain	Canonical value
Ω_{geo}	$1 / \phi_G$ (inverse geodesic length)	Geometry	$1/(2\pi)$ for SO(2)
Ω_{int}	Z_2 -intrinsic Bernoulli-unit precision	Z_2 rupture	1 by normalisation

In Z_2 -intrinsic units ($\Omega_{\text{int}} = 1$, C measured in Bernoulli draws), $\exp(C/\Omega) = e$ at the rupture $C \cdot \Omega = 1$.

In geometric units ($\Omega_{\text{geo}} = 1/\phi_G$), $\exp(C/\Omega_{\text{geo}}) = \exp(\phi_G^2)$ at the rupture, much larger than e for canonical phase manifolds ($e^{\pi^2} \approx 19,334$ for Z_2 ; $e^{4\pi^2} \approx 1.4 \times 10^{17}$ for SO(2)).

Convention adopted: unless otherwise noted, Ω in CRR formulas refers to Ω_{geo} . The brief's 'exp $\rightarrow$ e at rupture' framing is correct only in intrinsic units.

8. Coda — what CRR commits to, and what it does not

CRR commits to:

- Process is ontologically prior to substance (Axiom 1).
- Coherence accumulation is integration of Fisher-Rao squared speed (Axiom 2 + M13).
- Rupture is a Z_2 Bernoulli event (C1) at the Cramér-Rao saturation bound (Axiom 5 + M3).
- Regeneration is exponential weighting of past coherence (Axiom 4).
- Precision Ω is the inverse geodesic length of a compact connected Lie-group memory manifold (Axiom 6 + C2 + C4).
- CV scales as $1/(2 \cdot \phi_G)$ per Lie group (M22).

CRR does NOT commit to:

- Predicting CV = $1/(4\pi)$ for systems without an identified SO(2) phase manifold. Externally-driven event timing defaults to no-regulator CV ≈ 1 .
- Predicting Hawkes CV > 1 universally. The self-exciting regime requires explicit positive-feedback dynamics; physical event timing typically relaxes to equilibrium.
- Beating ETAS in California seismicity (P5 null). Single- Ω CRR matches; nested-CRR underperforms.
- Numerical agreement at full CODATA precision for the fine-structure fixed point ($1/\alpha = 137.0324$ vs CODATA 137.036, 26 ppm discrepancy at T1).

Per CAMPAIGN.md PART III: this canonical document does not promote any claim's tier. It records the post-Session-15 state of CRR's axiomatic and equational commitments, with corrections (M16, M19, Class C $\rightarrow$ no-regulator) integrated, relabellings (M5, M14) capped at T1*, and the CV ladder formalised. Tier promotions remain governed by per-claim evidence files in *claims/*.

— END OF CANONICAL REFERENCE —